

Khai Nguyen

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Education

Northeastern University, Boston, MA

Bachelor of Science in Mechanical Engineering and Computer Science

Expected Graduation: April 2030

Relevant Coursework: Introduction to Programming, Physics for Engineering I, Calculus I-III, Differential Equations and Linear Algebra for Engineering, Statistics, AP Computer Science A

Projects

Emberdeep | HTML, CSS, JavaScript

- Built Emberdeep, a browser-based roguelike dungeon crawler in vanilla JavaScript, HTML5 Canvas, and CSS—fully self-contained with zero external dependencies or build tools.
- Implemented procedural dungeon generation and BFS-based enemy AI, reusing the same graph-search algorithm for pathfinding and for validating that every randomly generated level is solvable (with automatic retries and a safe fallback).
- Wrote a headless Node.js test harness to stress-test dungeon generation and AI logic across hundreds of simulated runs, catching edge cases like unreachable exits and invalid spawns without a browser environment.
- **Github:** [Repository](#) | **Live Demo:** [Play Game](#)

Rubik's Cube Speed Timer | HTML, CSS, JavaScript

- Built and deployed a responsive speedcubing timer web app using vanilla HTML, CSS, and JavaScript, implementing a competition-style state machine and performance.now()/requestAnimationFrame for millisecond-accurate timing with zero external dependencies.
- Designed a WCA-style scramble generator and statistics engine (Ao5, Ao12, session averages) with correct handling of edge cases like DNF penalties and insufficient sample sizes, validated through 50+ automated unit and integration tests.
- Implemented client-side data persistence with localStorage, including defensive parsing to gracefully recover from corrupted or missing data, and deployed the app via GitHub Pages.
- **Github:** [Repository](#) | **Live Demo:** [Play Game](#)

Reaction Time Game | HTML, CSS, JavaScript

- Built and deployed an interactive browser-based reaction-time game featuring randomized timing, multiple difficulty levels, performance statistics, and persistent high scores using local storage.
- Developed game-state management and event-driven interactions in vanilla JavaScript to track user input and accurately measure reaction times.
- **Github:** [Repository](#) | **Live Demo:** [Play Game](#)

Skills

Technical: (*proficient*): Java, JavaScript, (*familiar*): Python

Language: Vietnamese